

Lesson 4: When Technology and Humanity Cross**Television Sets, Mobile Phones, Computers, and Humanity**

A number of technological devices can be easily found inside the home, the most accessible place to anyone. Having said that, it can also be easily inferred that these technological devices are some of the most popular and most commonly used types of devices across all age groups. Almost all households, if not all, own these types of devices. To be more specific, these "celebrities" in the field of technology are television sets, mobile phones, and computers. People all over the world use these technologies every day to accomplish different purposes.

First, according to Kantar Media, one of the most trusted television audience measurement providers, in the Philippines, 92 percent of urban homes and 70 percent of rural homes own at least one television set. It is for this reason why television remains to be the ultimate medium for advertisement placements (The Manila Times, 2014). This survey simply shows that almost all Filipinos use this particular type of device. In fact, Filipinos are believed to have this big fascination for television. Most of the time, they watch television during their free time or any time of the day when they have nothing important to do. In addition to this, Kantar Media also reported that in the Philippines, the current count of households with television set already reached 15.135 million (Noda, 2012). This number signifies something, that is, television plays a great role in the lives of the people or in this case, the Filipinos.

Television was a product of different experiments by various people. Paul Gottlieb Nipkow, a German student, in the late 1800s was Successful in his attempt to send images through wires with the aid of a rotating metal disk. This invention was then called the "electric telescope" that had 18 lines of resolution. After some time, in 1907, two inventors, Alan Archibald Campbell-Swinton who was an English scientist and Boris Rosing who was a Russian scientist, created a new system of television by using the cathode ray tube in addition to the mechanical scanner system. This success story gave rise to two types of television systems, namely, mechanical and electronic television (Jezek, n.d.). These experiments inspired other scientists to improve the previous inventions, which led to the modern television people now have. However, it is important to remember that several scientists and several experiments were performed first before finally achieving the modern television at present.

Second, Filipinos love to use their mobile phones anywhere, anytime. They use it for different purposes other than for communication. More than half of the Filipino population own at least one mobile phone regardless of type. In 2010, global research agency

Synovate conducted a survey and declared 67 percent product ownership in the country. In fact, it was also claimed that mobile phones are considered a must-have among young Filipinos (ABS-CBN News, 2010). To prove that Filipinos really love to use their mobile phones, the Ipsos Media Atlas Philippines Nationwide Urban 2011-2012 survey results showed that one in every three Filipinos cannot live without a mobile phone. In other words, 30 percent of the Philippine urban population nationwide said that mobile phones are necessities in life (Roa, 2012). Philippine streets are full of people using their mobile phones. Not only this, there are some Filipinos who even own more than one mobile phone.

Mobile phones have a very interesting background story. On April 3, 1973, Martin Cooper, a senior engineer at Motorola, made the world's mobile phone call. He called their rival telecommunications company first properly informed them that he was making the call from a mobile phone. The mobile phone used by Cooper weighed 1.1 kilograms and measured 228.6 x 127 x 44.4 mm. This kind of device was capable of a 30-minute talk time. However, it took 10 hours to charge. In 1983, Motorola made their first commercial mobile phone available to the public. It was known as the Motorola DynaTAC 8000X (Goodwin, 2016).

Lastly, computers and laptops have also become part of many of the Filipino households. There are some Filipino families who own more than one computer or laptop while some own at least one computer or laptop. However, the number of computers or laptops sold per year may not be as high as the number of mobile phones and television sets. This is because of the relatively higher cost of computers and laptops. While it is true that almost all Filipino families own at least one television set and a mobile phone, it is not possible for all Filipino families to own at least one computer or laptop. In fact, most of the profits gained by computer and laptop manufacturers come from offices, businesses, or schools where such devices have become part of their necessities.

In 2010, 3.6 trillion was the estimated total value output of all manufacturing establishments. Semi-conductor devices and other electronic components took more than half of the total value output of all manufacturing establishments. To be more specific, 5.4 percent of the total value output came from computers and peripheral equipment and accessories (PSA, 2013). In line with the growing number of computer and laptop sales, there have also been a growing number of Internet users in the Philippines. Although there have been problems regarding the Internet providers, this never hindered Filipinos from continuously using the Internet.

Just like television sets and mobile phones, computers and laptops also have a long background history of trial and error. It was Charles Babbage, a nineteenth-century English Mathematics professor, who designed the Analytical Engine which was used as the basic framework of the computers even until the present time. In general,

computers can be classified into three generations. Each generation of the computer was used for a certain period of time and each gave people a new and improved version of the previous one (Steitz, n.d.).

Laptops have been available to the public for even less time than personal computers. Before, the first design of computer was so big that it could occupy whole floors of buildings. It was not long before people started dreaming that they could bring their devices to any place they wished. They hoped that someday it would be possible for these devices to be portable. It was believed that the transition from a personal computer to a laptop was only a matter of design, an improvement and a little deviation from the standard design of a personal computer. The first true portable computer was released in April 1981. It was called the Osborne 1 (Orfano, 2011). From that moment on, the evolution of laptops continued until the present time where various designs and models are already available.

A typical household owns at least four of the following devices: a mobile phone (89%), smartphone (53%), tablet (14%), desktop (39%), laptop or netbooks (37%), and smart TV (4%) (Philstar, 2013). These data prove the deep-seated fascination of Filipinos to different technological devices.

Here are some facts about Filipinos and their use of gadgets and the Internet (Rappler, n.d.):

- Mobile phone subscription is at 119 million.
- Filipinos spend approximately 3.2 hours on mobile and 5.2 hours on desktop daily.
- Currently, the Philippines have one of the highest digital populations in the world.
- There are now 47 million active Facebook accounts in the Philippines.
- The Philippines is the fastest-growing application market in Southeast Asia.

Roles Played by These Technological Advancements

Television sets, mobile phones, and computers or laptops all have different functions and roles played in the lives of the people, although may be a little similar. These roles have become so essential that people, more specifically Filipinos, developed a strong inclination toward technology and its products.

For instance, television is mainly used as a platform for advertisements and information dissemination. In fact, television remains to be the most used avenue by different advertising companies not only in the Philippines but also all over the world. Various advertising companies trust that television is still one of the most used technological devices up until today. However, it is important to note that these are just some of the roles played by television. Aside from the ones mentioned above, it also serves as a recreational activity and good stress reliever to most families, specifically to Filipino families. Television also is a good platform for different propagandas and advocacies. Lastly, it can also be a good way to bond with one's family members.

Mobile phones, on the other hand, also have their own roles in the lives of the people. They are primarily used for communication. Mobile phones offer services like texting and calling. In the past, these were the only functions of the mobile phone but as technology progressed, there have been many additional features included on mobile phones. For example, in the present, people use their mobile phones to surf the Internet and to take pictures more than to text or to call people. This is the reason why more and more people all over the world prefer to buy smartphones over the old models where such features are not available. Other applications include music player, calendar, radio, television, and photo editor, among others. These are just some of the additional features of mobile phones in the present. These make this particular technological device very appealing to the masses. It is like an all-in-one device. In addition, it is very portable and convenient because it can fit into any space, may it be inside the pocket or bag.

Personal computers and laptops also have useful set of functions and roles. Although most of the functions found in these technological devices are now also available in mobile phones, they still offer their own unique features that make them attractive. For example, personal computers and laptops can be used to surf the Internet and communicate. Just like the mobile phones, these devices also have features like calendar, calculator, music player, movie player, camera, and many more. However, for a lot of people, they prefer to do their job using either a personal computer or a laptop than a mobile phone. One reason is that a personal computer or a laptop has wide keyboard than using a mobile phone, especially when the mobile phone has a small screen. Contrary to mobile phones, personal computers and laptops have wide screens and separate keyboards, although some mobile phones can now be connected to a keyboard. Another reason is that the availability of a mouse or a touchpad made these two technological devices easier to maneuver than mobile phones. Lastly, for the youth and those who love to play different computer games, personal computers or laptops are really the better choice because these allow them to play with comfort and convenience. However, it cannot be denied that some would even prefer laptops over personal computers for the simple reason that personal

computers are not portable and there are times when they need to bring such devices to different places.

Ethical Dilemma Faced by These Technological Advancements

While it is true that these technological devices are useful and beneficial, the fact remains that there are several dilemmas faced by these "necessities." First, most parents would argue that these devices make their children lazy and unhealthy. This is because of the fact that people who are fixated on these technological advancements start and end their day by using such devices. They have a great tendency to sit and chill all day long without doing anything productive in their homes, thus making them unhealthy because they do not just skip meals sometimes but also lack exercise or any bodily movements. Here, it can be inferred that these types of people are already overly dependent on these technological devices for example, those who love to watch television shows stay in front of the television for more than six hours a day while those who love to surf the Internet or play computer games stay on their laptops, computers, or mobile phones for more than half a day. These people have the tendency to be unaware of the time because they are so engrossed with the use of technological device. In fact, if they get disturbed, there is a great chance that they will get mad or annoyed. Moreover, these are the same people who are more likely to experience alienation because they no longer take the time to get out of their houses and mingle with other people.

Another dilemma faced by these technological devices is the moral dilemma. People, especially the children who are not capable yet of rationally deciding for themselves what is right or wrong, are freely exposed to different things on television, mobile phones, laptops, or computers. Because of the availability and easy access to the Internet, they can just easily search the web and go to different websites without restrictions. This allows them to see, read, or hear things which are not suitable for their very young age. This makes them very vulnerable to character change and can greatly affect the way they view the world and the things around them.

On the first dilemma, it is really concerning to know that there are people who develop different kinds of sickness because of too much use of technological devices. Not only this, it also causes them to become reclusive, alienating themselves from other people. Although some would argue that technology brings people together, it can also be argued that this is not always the case in the real world because it may bring them virtually closer but not physically or personally. In fact, there are people who are friends, for example, only on social media but not in real life. This just shows that there are things that technology claims to do but in reality, does not. It is for these reasons why there are people who call for the establishment of ethics of technology. This subcategory of ethics

will in one way or another guide people on how technology ought to be used in order to prevent abuse and other unfortunate results.

Digging deeper, it can be said that these reasons make such devices somewhat unethical because they bring undesirable consequences to people. However, it can also be argued that it is not the fault of the technological devices but the agents using them or the ones making them. The classic deontological and teleological approaches to ethics are already too old to be applied in such cases. This is because technology has become very complicated and dynamic over time. Having said this, it is true that there are problems that can no longer be addressed by using these theories only. This is why the ethics of responsibility is an appropriate theory that can be used in these dilemmas.

The word "responsibility" in the sense of being accountable for and St: accountable to is very appropriate to the ethics of technology because it ye makes each and every person in the scientific-technological development a proxy with reference to one another. In other words, each person must indicate the priorities, values, norms, and principles that constitute the grounds for one's actions and define one's contribution to the scientific-technological event. The ethics of responsibility focuses on the positive rather than the negative. Instead of asking "What ought not to be allowed?" ask "What ought to be allowed?" To put it in another way, people who are part of the scientific development ought to let the public know the good in their respective technological contribution/s. In this way, the people will have an idea how the devices ought to be used in order to maximize their positive results.

However, it is also important for the people in the scientific world to inform the masses of the dangers of their contribution/s to the world of technology. In this way, the people will be sufficiently aware of what to do and what not to do. In addition to this, the agents using the devices should also be accountable to and accountable for their use of their gadgets.

Going back to the first dilemma, it can be said that the agents using the devices are the ones to be blamed for the undesirable consequences, namely, laziness and unhealthiness. However, it is the assumption that the people in the scientific-technological world have properly informed the public of the positive tenor of their action in technology and the possible dangers of the misuse of their technological contribution. Thus, the undesirable consequences are brought about by the misuse of the agent. Now, talking about alienation, it can be concluded that the people that the scientific-technological world are blameworthy because they tell the people something that seems positive but when examined closely, brings more bad than good.

On the second dilemma, the people in the scientific world nor the children are blameworthy because first, the children are not yet capable of rationally deciding for themselves what is good and what is bad. Second, even if creators of these technologies went out of their way to inform children of the pros and cons of these technological contributions, it would be useless because the children have no capacity to understand them yet. So in this dilemma, the ones to be blamed are the adults who allowed children to have access to such devices in the first place without any supervision. It is the recklessness and overconfidence of the adults that cause the character change in children.

Robotics and Humanity

Another great product of the innovative minds of the people is the robot. Robots are now widely used. For example, there are the so-called service robots. These particular robots do specific tasks but focus mainly in assisting their masters in their everyday tasks. The International Federation of Robotics (IFR) and United Nations Economic Commission for Europe (UNECE) made it their task to formulate a working definition for service robots. A preliminary extract of the relevant definition is (IFR, 2012):

- A robot is an actuated mechanism programmable in two or more axes with a degree of autonomy, moving within its environment, to perform intended tasks. Autonomy in this context means the ability to perform intended tasks based on current state and sensing without human intervention.
- A **service robot** is a robot that performs useful tasks for humans or equipment excluding industrial automation application. Note: A robot may be classified according to its intended application as an industrial robot or a service robot.
- A **personal service robot or a service robot** for personal use is a service robot used for a noncommercial task, usually by laypersons. Examples are domestic servant robot, automated wheelchair, personal mobility assist robot, and pet exercising robot.
- A **professional service robot or a service robot for professional use** is a service robot used for a commercial task, usually operated by a properly trained operator. Examples are cleaning robot for public places, delivery robot in offices or hospitals, fire-fighting robot, rehabilitation robot, and surgery robot in hospitals. In this context, an operator is a person designated to start, monitor, and stop the intended operation of a robot or a robot system.

Germany was one of the first countries to develop service robots. As part of the German Federal Ministry of Education and Research's "Service Robotics Innovation Lead Initiative," it sponsored a collaborative project called DESIRE (Deutsche Servicerobotik

Initiative-Germany Service Robotics Initiative) which was launched on October 1, 2005. DESIRE has the following individual objectives (DESIRE, 2009):

- To achieve a technological edge toward attaining key functions and components that are suited for everyday use
- To create a reference architecture for mobile manipulation
- To promote the convergence of technologies through integration into a common technology platform
- To conduct pre-competition research and development activities for new products and technology transfer in start-up enterprises in the field of service robotics

Some of the expected work to be performed by DESIRE are the following: (1) "Clear up the kitchen table" - all objects on top of the kitchen table will be moved to where they belong; (2) "Fill the dishwasher" - the dirty dishes will be sorted correctly into the dishwasher; and (3) "Clear up this room" - all objects that are not in their proper places will be moved to where they belong (Mock, n.d.).

The earliest conception of robots can be traced around 3000 B.C. from the Egyptians. Their water clocks used human figurines to strike the hour bells. This mechanical device was built to carry out a specific physical task regularly. From that time on, different machines were already built that displayed the same mechanism and characteristics as the robots in the present. For example, there was a wooden pigeon that could fly, a talking doll, steam-powered robots, and hydraulically-operated statues that could speak and gesture. However, the earliest robots as people know them were created in the early 1950s by George Devol. "Unimate" was his first invention from the words "Universal Automation." Unfortunately, his attempt to sell his product to the industry did not succeed. After Unimate, several robots were also invented which were better versions of the previous ones (Stanford, n.d.). Ever since, people never stopped their quest in the field of robotics.

Roles Played by Robotics

Robots play different roles not only in the lives of the people but also in the society as a whole. They are primarily used to ease the workload of mankind. They were invented to make life more efficient and less stressful. On one hand, they perform complicated activities which human beings are incapable of doing. On the other hand, they perform the simplest tasks at home so that their masters can perform the complex ones without stressing themselves over the simple tasks. There are also robots which are made for pleasure. To be more specific, these types of robots perform activities to entertain people. They can usually be found in amusement parks or exhibits. In addition, there are also some robots which were made to serve as toys. They also perform different activities but

they are usually child- friendly. Other examples of robots are those which can be seen in movies. One of the reasons why robots are very famous is because of movies. A number of local and national movies were inspired by robots. This goes to show that people have developed a distinct fascination over robots.

Just like people living in the society, robots also have their own set of rules and characteristics that define what a good robot is. These laws were formulated by Isaac Asimov back in the 1940s, when he was thinking of the ethical consequences of robots. These are the following (Stanford, n.d.):

Law One:

A robot may not injure a human being or, through inaction, allow a human being to come to harm.

Law Two:

A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.

Law Three:

A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Ethical Dilemma/s Faced by Robotics

Just like any other technological advancements, robotics also faces different problems and dilemmas. Although the idea is to help people and make their lives a lot easier than before, it is still not immune to different ethical dilemmas and possible undesirable outcomes. One of the dilemmas faced by robots is safety. Who should be held accountable if someone's safety is compromised by a robot? Who should be blamed, the robot, the agent using the robot, or the maker/inventor of the robot? It is important to know who should be blamed and who should be held responsible if such thing happens.

Another ethical dilemma faced by robots is the emotional Component. This may seem a little absurd as of the moment, but looking at how fast technology progresses nowadays, it is not completely impossible for robots to develop emotions (Evans, 2007). So here, the questions become, "What if robots become sentient? Should they be granted robot rights? Should they have their own set of rights to be upheld, respected, and protected by humans?" It is interesting to know how people would react if the time comes when robots can already feel pain and pleasure. Would they act differently or not at all?

In the field of robotics, there are the so-called partial autonomy and full autonomy. Partial autonomy includes active human-robot interaction while full autonomy excludes active human-robot interaction. In Words, a robot with full autonomy can perform actions or activities even without a master telling it what should be done or what should be performed next (IFR, 2012).

Using Asimov's laws for robots, it can be concluded that robots are ethical but only if they strictly follow the laws specified. They are tr ethical mostly because the laws formulated by Asimov ensure the safety a of not only the users of the technology but also the people around him. Remember that these service robots are already available to the public; thus, they can already be found inside the homes. Having said that, the safety of not only the owner of the technology but also all the people inside the house should be the priority more than anything else. In other words, the service robots only follow what their masters tell them to do all with great consideration to the laws formulated by Asimov. However, if the agent using the technology misuses the robot to achieve personal agendas, then without a doubt, the agent should be held accountable for any consequences it may bring. It is important to note that this is under the assumption that the robot strictly followed the laws specified without any form of deviation.

If the problems arise when the robot deviates from the laws specified, then the maker or the inventor of the machine should be blameworthy. It just means that the robot was not programmed very well because it violated the laws. Other problems may arise when the machine develops for the ability to think for itself. In this case, the one that should be blame can both be the maker or inventor and the robot itself. This is because, in day, the first place, the maker gave the robot the capacity to think for itself so he should be very much aware of its possible consequences. To put it in another way, the maker programmed the robot in such a way that it can already think for itself even without an active participation from a human being. In addition, since the robot thinks for itself, whatever decision it makes and whatever consequence it may bring, the robot itself should be held responsible.

For the second dilemma, it is just right for the robots to be given their own set of rights should they develop the ability to feel different kinds of emotion. It can be argued that the same thing happened with animals. Before, animals did not have their own set of rights because people believed that they were not capable of having emotions. However, after years of testing and experimenting, it was concluded that animals are indeed capable of emotions. It is for this reason that people decided to give them rights that are due to them. The same should be done to robots without any reservations. Should that time come, they ought to be treated differently and they ought to have new laws to follow in order to accommodate the new characteristic they have developed.

SPECIFIC ISSUES IN SCIENCE, TECHNOLOGY, AND SOCIETY

Lesson 1: The information Age

Highly modernized, automated, data-driven, and technologically advanced-these best describe our society nowadays, as evidenced by how information could be transferred or shared quickly. The different areas of society have been influenced tremendously such as communication, economics, industry, health, and the environment. Despite our gains due to the growing development of information technology, the rapid upgrade of information also has disadvantages. This lesson will discuss the history and impact of technological advancements to society.

Life is accompanied by endless transmission of information that takes place within and outside the human body. According to Webster's Encyclopedic Unabridged Dictionary, information is "knowledge communicated or obtained concerning a specific fact or circumstance." Hence, information is a very important tool for survival.

The Information Age is defined as a "period starting in the last quarter of the 20th century when information became effortlessly accessible through publications and through the management of information by computers and computer networks" (Vocabulary.com, n.d.). The means of conveying symbolic information (e.g., writing, math, other codes) among humans has evolved with increasing speed. The Information Age is also called the Digital Age and the New Media Age because it was associated with the development of computers.

According to **James R. Messenger** who proposed the Theory of Information Age in 1982, "the Information Age is a true new age based upon the interconnection of computers via telecommunications, with these information systems operating on both a real-time and as-needed basis. Furthermore, the primary factors driving this new age forward are convenience and user-friendliness which, in turn, will create user dependence."

History

The table below traces the history and emergence of the Information Age.

Year	Event
3000 BC	Sumerian writing system used pictographs to represent words
2900 BC	Beginnings of Egyptians hieroglyphic writing
1300 BC	Tortoise shell and oracle bone writing were used
500 BC	Chinese small seal writing was developed
100 AD	Book (parchment codex)
105 AD	Woodblock printing and paper was invented by the Chinese
1455	Johannes Gutenberg invented the printing press using movable metal type
1755	Samuel Johnson's dictionary standardized English spelling
1802	The library of Congress was established Invention of the carbon arc lamp
1824	Research on persistence of vision published
1830s	First viable design for a digital computer Augusta Lady Byron writes the world's first computer program
1837	Invention of the telegraph in Great Britain and the United States
1861	Motion pictures were projected onto a screen
1876	Dewey Decimal system was introduced
1877	Eadweard Muybridge demonstrated high-speed photography
1899	First magnetic recordings were released
1902	Motion picture special effects were used

1906	Lee DeForest invented the electronic amplifying tube (triode)
1923	Television camera tube was invented by Zvorkyn
1926	First practical sound movie
1939	Regularly scheduled television broadcasting began in the US
1940s	Beginning of information science as a discipline
1945	Vannevar Bush foresaw the invention of hypertext
1946	ENIAC computer was developed
1948	Birth of field-of-information theory proposed by Claude E. Shannon
1957	Planar transistor was developed by Jean Hoerni
1958	First integrated circuit
1960s	Library of Congress developed LC MARC (machine-readable code)
1969	UNIX operating system was developed, which could handle multitasking
1971	Intel introduced the first microprocessor chip
1972	Optical laserdisc was developed by Philips and MCA
1974	MCA and Philips agreed on a standard videodisc encoding format
1975	Altair Microcomputer Kit was released: First personal computer for the public
1977	RadioShack introduced the first complete personal computer
Mid 1980s	Artificial Intelligence was separated from information science
1987	Hypercard was developed by Bill Atkinson recipe box metaphor
1991	Four hundred fifty complete works of literature on one CD ROM

	was released
January 1997	RSA (encryption and network security software) Internet security code cracked for a 48-bit number

As man evolved, information and its dissemination has also evolved in many ways. Eventually, we no longer kept them to ourselves; instead, we share them and manage them in different means. Information got ahead of us. It started to grow at a rate we were unprepared to handle. Because of the abundance of information, it was difficult to collect and manage them starting in the 1960s and 1970s. During the 1980s, real angst set in. Richard Wurman called it "Information Anxiety." In the 1990s, information became the currency in the business world. Information was the preferred medium of exchange and the information managers served as information officers. In the present generation, there is no doubt that information has turned out to be a commodity, an overdeveloped product, mass-produced, and unspecialized. Soon, we become overloaded with it.

Different authors have diverse, contrasting ideas on the evolution of the Information Age. In spite of this, we can still say that information is a very important tool that helps improve our way of life. One thing is for sure, the Information Age will continue to move forward and far greater than our minds could imagine.

In his article "Truths of the Information Age" (n.d.), Robert Harris detailed some facts on the Information Age.

1. Information must compete. There is a need for information to stand out and be recognized in the increasing clutter.
2. Newer is equated with truer. We forgot the truth that any fact or value can endure.
3. Selection is a viewpoint. Choose multiple sources for your information if you want to receive a more balanced view of reality.
4. The media sells what the culture buys. In other words, information is driven by cultural priorities.
5. The early word gets the perm. The first media channel to expose an issue often defines the context, terms, and attitudes surrounding it.
6. You are what you eat and so is your brain. Do not draw conclusions unless all ideas and information are presented to you.

7. Anything in great demand will be counterfeited. The demand for incredible knowledge, scandals, and secrets is ever-present; hence, many events are fabricated by tabloids, publicists, or other agents of information fraud.
8. Ideas are seen as controversial. It is almost certainly impossible to make any assertion that will not find some supporters and some detractors.
9. Undead information walks ever on. Rumors, lies, disinformation, and gossips never truly die down. They persist and continue to circulate.
10. Media presence creates the story. People behave much differently from the way they would if being filmed when the media are present, especially film news or television media.
11. The medium selects the message. Television is mainly pictorial, partially aural, and slightly textual, so visual stories are emphasized: fires, chases, and disasters.
12. The whole truth is a pursuit. The information that reaches us is usually selected, verbally charged, filtered, slanted, and sometimes, fabricated. What is neglected is often even more important than what is included.

Computer

Computers are among the most important contributions of advances in the Information Age to society. A computer is an electronic device that stores and processes data (information). It runs on a program that contains the exact, step-by-step directions to solve a problem (USHistory. org, 2017).

Types of Computer

Computers are associated with numerous terms and descriptions. Most people suggest the dimensions, intended use, or the computer's power. While the term "computer" can apply to virtually any device that has a microprocessor in it, most people think of a computer as a device that receives input from the user through a mouse (hand-guided directions tool) or keyboard, processes it in some fashion, and presents the result on a screen.

1. Personal Computer (PC)

It is a single-user instrument. PCs were first known as microcomputers since they were a complete computer but built on a smaller scale than the enormous systems operated by most businesses.

2. Desktop Computer

It is described as a PC that is not designed for portability. The assumption with a desktop is that it will be set up in a permanent spot. A workstation is simply a desktop computer that has a more powerful processor, additional memory, and enhanced capabilities for performing special group of tasks, such as 3D graphics or game development. Most desktops offer more storage, power, and versatility than their portable versions (UShistory.org, 2017).

3. Laptops

These are portable computers that integrate the essentials of a desktop computer in a battery-powered package, which are somewhat larger than a typical hardcover book. They are commonly called notebooks.

4. Personal Digital Assistants (PDAs)

These are tightly integrated computers that usually have no keyboards but rely on a touch screen for user input. PDAs are typically smaller than a paperback, lightweight, and battery-powered (UShistory.org, 2017).

5. Server

It refers to a computer that has been improved to provide network services to other computers. Servers usually boast powerful processors, tons of memory, and large hard drives (UShistory.org, 2017).

6. Mainframes

These are huge computer systems that can fill an entire room. They are used especially by large firms to describe the large, expensive machines that process millions of transactions every day. The term "mainframe" has been replaced by enterprise server. Although some supercomputers are single computer systems, most comprise multiple, high-performance, parallel computers working as a single system.

7. Wearable Computers

They involve materials that are usually integrated into cell phones, watches, and other small objects or places. They perform common computer applications such as databases, email, multimedia, and schedulers.

The World Wide Web (Internet)

Several historians trace the origin of the Internet to Claude E. Shannon, an American Mathematician who was considered as the "Father of Information Theory." He worked at Bell Laboratories and at age 32, he published a paper proposing that information can be quantitatively encoded as a sequence of ones and zeroes.

The Internet is a worldwide system of interconnected networks that facilitate data transmission among innumerable computers. It was developed during the 1970s by the Department of Defense. In case of an attack, military advisers suggested the advantage of being able to operate on one computer from another terminal. In the early days, the Internet was used mainly by scientists to communicate with other scientists. The Internet remained under government control until 1984 (Rouse, 2014).

One early problem faced by Internet users was speed. Phone lines could only transmit information at a limited rate. The development of fiber-optic cables allowed for billions of bits of information to be received every minute. Companies like Intel developed faster microprocessors so personal computers could process the incoming signals at a more rapid rate (USHistory.org, 2017).

Sergey Brin and Larry Page, directors of a Stanford research project, built a search engine that listed results to reflect page popularity when they determined that the most popular result would frequently be the most usable. After talking with family, friends, and other investors into contributing \$1 million, the researchers launched their company in 1998. Google is now the world's most popular search engine, accepting more than 200 million queries daily.

Back then, new forms of communication were also introduced. Electronic mail, or email, was a suitable way to send a message to fellow workers, business partners, or friends. Messages could be sent and received at the convenience of the individual. A letter that took several days to arrive could be read in minutes. Internet service providers like America Online and CompuServe set up electronic chat rooms. These were open areas of cyberspace where interested parties could join in a conversation with perfect strangers. "Surfing the net" became a pastime in and of itself.

Consequently, companies whose businesses are built on digitized information have become valuable and powerful in a relatively short period of time; the current Information Age has spawned its own breed of wealthy influential brokers, from Microsoft's Bill Gates to Apple's Steve Jobs to Facebook's Mark Zuckerberg.

Critics charged that the Internet created a technological divide that increased the gap between the members of the higher class and lower class of society. Those who could not

afford a computer or a monthly access fee were denied these possibilities. Many decried the impersonal nature of electronic communication compared to a telephone call or a handwritten letter.

On one hand, the unregulated and loose nature of the Internet allowed pornography to be broadcast to millions of homes. Protecting children from these influences or even from meeting violent predators would prove to be difficult. Nowadays, crimes in various forms are rampant because of the use of social media. Cyber bullying is an issue that poses alarm worldwide. Consequently, we need to be aware of the possible harm and damage due to abuse of these advances in the Information Age.

Applications of Computers in Science and Research

One of the significant applications of computers for science and research is evident in the field of bioinformatics. Bioinformatics is the application of information technology to store, organize, and analyze vast amount of biological data which is available in the form of sequences and structures of proteins-the building blocks of organisms and nucleic acids-the information carrier (Madan, n.d.). Early interest in bioinformatics was established because of a need to create databases of biological sequences. The human brain cannot store all the genetic sequences of organisms and this huge amount of data can only be stored, analysed, and be used efficiently with the use of computers.

While the initial databases of protein sequences were maintained at individual laboratories, the development of a consolidated formal database, known as **SWISSPROT** protein sequence database, was initiated in 1986. It now has about 70,000 protein sequences from more than 5,000 model organisms, a small fraction of all known organisms. The enormous variety of divergent data resources is now available for study and research by both academic institutions and industries. These are made available as public domain information in the larger interest of research community through the Internet and CD-ROMs. These databases are constantly updated with additional entries (Madan, n.d.).

Computers and software tools are widely used for generating these databases and to identify the function of proteins, model the structure of proteins, determine the coding (useful) regions of nucleic acid sequences, find suitable drug compounds from a large pool, and optimize the drug development process by predicting possible targets. Some of the software tools which are handy in the analysis include: BLAST (used for comparing sequences); Annotator (an interactive genome analysis tool); and GeneFinder (tool to identify coding regions and splice sites) (Madan, n.d.).

The sequence information generated by the human genome research, initiated in 1988, has now been stored as a primary information source for future applications in medicine.

The available data is so huge that if compiled in books, the data would run into 200 volumes of 1,000 pages each and reading alone (ignoring understanding factor) would require 26 years working around the clock. For a population of about five billion human beings with two individuals differing in three million bases, the genomic sequence difference database would have about 15,000,000 billion entries. The present challenge to handle such huge volume of data is to improve database design, develop software for database access, and manipulation and device data-entry procedures to compensate for the varied computer procedures and systems used in different laboratories. The much-celebrated complete human genome sequence which was formally announced on the 26th of June 2000 involved more than 500×10^{18} (500 million trillion) calculations during the process of assembling the sequences alone. This can be considered as the biggest exercise in the history of computational biology (Madan, n.d.).

Examples of Useful and Reliable Web Sources

1. **AFA e-Newsletter** (Alzheimer's Foundation of America newsletter)
2. **American Memory the Library of Congress historical digital collection.**
3. **Bartleby.com Great Books Online** - a collection of free e-books including fictions, nonfictions, references, and verses.
4. **Chronicling America search and view pages from American newspapers from 1880-1922.**
5. **Cyber Bullying a free collection of e-books from ebrary** plus additional reports and documents to help better understand, prevent and take action against this growing concern.
6. Drug information websites:
 - ↗ **National Library of Medicine's MedlinePlus** ↗ **Drugs.com**
 - ↗ **PDRhealth**
7. **Global Gateway:** World Culture & Resources (from the 7. Library of Congress)
8. **Google Books**
9. **Googlescholar.com**
10. History sites with primary documents:
 - **AMDOCS:** Documents for the study of American history

- **Avalon Project:** Documents in Law, History and Diplomacy (Yale Law School)
 - **Internet Modern History Sourcebook:** Colonial Latin America
 - **Teacher Oz's Kingdom of History**
11. **Illinois Digital Archives** - the Illinois State Library working with libraries, museums, and historical societies in Illinois provides this collection of materials related to Illinois history.
 12. **Internet Archive** - a digital library of Internet sites and other cultural artifacts in digital form.
 13. **Internet Archive for CARLI digitized resources**
 14. **Internet Public Library**
 15. **ipl2**- a merger of Librarians' Internet Index and Internet Public Library. Special interest may include the "Literary Criticisms" page which can be found after clicking on the "Special Collections" link.
 16. **Librarians' Internet Index**
 17. Making of America a **digital library** of primary sources in American social history.
 18. **Maps**- from the University of Texas at Austin collection. Includes historical and thematic maps.
 19. **NationMaster** - a massive central data source and a handy way to graphically compare nations. It is a vast compilation of data from such sources as the CIA World Factbook, UN, and OECD.
 20. Nursing sites:
 - AHRQ
 - National Guidelines
 - Clearinghouse PubMed
 21. **Project Gutenberg**- the first and largest single collection of free electronic books with currently over 20,000 e-books available.
 22. **Shmoop** - literature, US history, and poetry information written primarily by PhD and masters students from top universities like Stanford, Berkeley, Harvard, and Yale.
 23. **StateMaster**- a unique statistical database which allows you to research and compare a multitude of different data on US states using various primary sources such

as the US Census Bureau, the FBI, and the National Center for Educational Statistics. It uses visualization technology like pie charts, maps, graphs, and scatter plots to provide data.

24. **Virtual Reference**- selected web resources compiled by the Library of Congress. One can also visit the university library and seek help from librarians as they are knowledgeable and the library has a rich collection of online library resources that are very useful for academic and research purposes.

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